Data Mining II 2024/2025

SHAPING THE FUTURE OF COLORECTAL CANCER SURVIVAL BY ADVANCING INSIGHTS TODAY

GROUP 05:

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Abstract

This project focuses on developing a binary classification model to predict the survival of colorectal cancer patients using a dataset comprising clinical, demographic, and lifestyle features.

The process began with extensive exploratory data analysis and preprocessing to uncover patterns and address data quality issues. Key steps included handling missing values and outliers, correcting formatting inconsistencies, engineering relevant features, and scaling numeric variables. Redundant and non-informative features were removed to reduce noise and improve model robustness.

Feature selection techniques, including Sequential Feature Selection and Recursive Feature Elimination with Cross-Validation, were used to identify the most predictive variables and reduce dimensionality. With these refined features, multiple machine learning algorithms – such as Logistic Regression, Random Forest, among others – were trained and evaluated using stratified 5-fold cross-validation. To enhance the generalization of the best-performing model on unseen data, GridSearch was applied for hyperparameter tuning and a Voting Classifier was used to surpass the performance of the individual model.

Final predictions were generated on the test set and submitted to a Kaggle competition for external evaluation. The final prediction output classified 42,734 patients as survivors and 32,266 as non-survivors, indicating a slight leaning toward positive classifications while remaining relatively balanced considering the class distribution.

These results suggest that while the model can identify some survival patterns, the performance remains modest – likely due to class overlap. Nevertheless, the current approach provides a meaningful starting point for automated survival prediction in CRC patients, with room for future refinement and clinical integration.

Introduction

Colorectal cancer (CRC) remains a significant global health burden, ranking as the third most diagnosed cancer worldwide and the second leading cause of cancer-related deaths. In 2022, approximately 1.9 million new cases were reported globally, accounting for 10% of all cancer diagnoses, while about 930,000 deaths (9.3% of all cancer-related fatalities) were attributed to CRC.

This project focuses on developing a predictive classification model to estimate whether a patient diagnosed with CRC will survive. The approach begins with thorough data preprocessing and exploratory data analysis (EDA) to uncover relevant patterns and correlations. Subsequently, the focus shifts to building and evaluating binary classification models, using a robust assessment strategy to compare, optimize, and select the most generalizable model.

Recent years have seen significant research into predictive models and survival analysis for CRC, with a focus on identifying key prognostic factors and improving patient outcomes. Numerous studies have applied machine learning algorithms to predict survival in CRC patients:

i. A 2025 study highlighted the effectiveness of models such as decision trees, random forest, support vector machines, and XGBoost. This last one, when combined with recursive feature elimination, achieved an accuracy of 0.83 and an AUC (area under the curve) of 0.89. This study identified perioperative transfusion quantity, tumor stage, carcinoembryonic antigen, CA19-9 and CA125[1] and patient age as major predictors.

ii. Traditional methods such as the Cox Proportional Hazards Model remain widely used. Studies have shown that factors such as pathological stage, surgery, health insurance status, age, residency and tumor site affect significantly survival.

The reviewed studies suggest that predictive models for CRC survival can achieve strong performance, especially when they incorporate a combination of clinical, demographic, and biomarker data. Notably, variables like tumor stage, age and treatment type consistently appear as the most influential predictors. These variables, which will be further discussed in the following section, are also present in the model's training dataset.

Data Exploration and Preprocessing

The dataset used in this project consists of patient records containing attributes related to demographics (gender, country of origin…), medical history, lifestyle factors, healthcare metrics (healthcare costs, insurance costs…) and lifestyle indicators (smoking history, alcohol consumption…). The target variable for the classification model is the survival prediction, indicating the likelihood of patient survival.

Exploratory data analysis

The initial data analysis shows that the dataset contains twenty-four categorical variables and five numerical ones, in addition to the target variable. This exercise focused on uncovering the statistical properties of the features and evaluating their predictive potential for the construction of classification models.

Regarding the first group of variables, the distribution analysis reveals that the variables 'Genetic Mutation', 'Heart Disease History', and 'Inflammatory Bowel Disease' exhibit strong imbalance, with at least 85% of the observations belonging to the majority class.

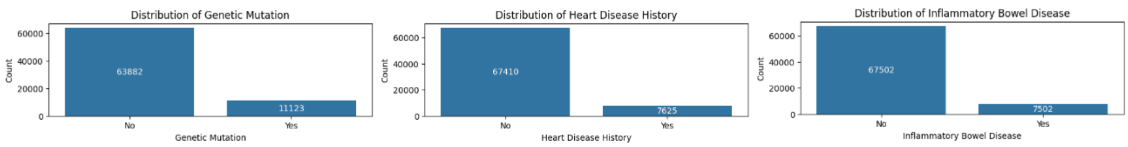

Figure 1 – Distribution of highly imbalanced features

On the other hand, it was found that the variables 'Transfusion History' and 'Diabetes History' have constant values, so they do not provide useful information for detecting patterns in the dataset.

[1] CA 19-9 and CA 125 are blood tests used to help detect and monitor certain types of cancer

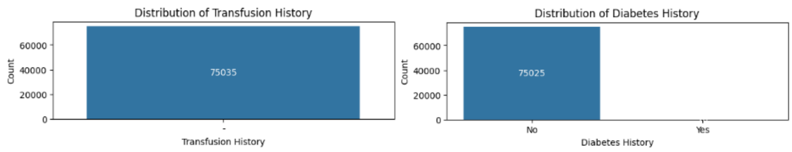

Figure 2 – Distribution of constant features

Other categorical variables like ‘Urban or Rural’ presented inconsistencies in formatting (more specifically, capitalization inconsistencies), which need to be standardized to ensure consistency.

Moreover, a chi-squared analysis of the categorical variables was performed to assess the degree of dependence among the features. This analysis revealed a strong relationship between ‘Non Smoker’ and ‘Smoking History’, indicating redundancy. Consequently, and given its lower association with the target variable, the ‘Smoking History’ variable will be dropped.

Variable 1	Variable 2	Chi2	p-value	DOF
Non Smoker	Smoking History	74202.841021	0.000000	1

Figure 3 – Chi-squared analysis results

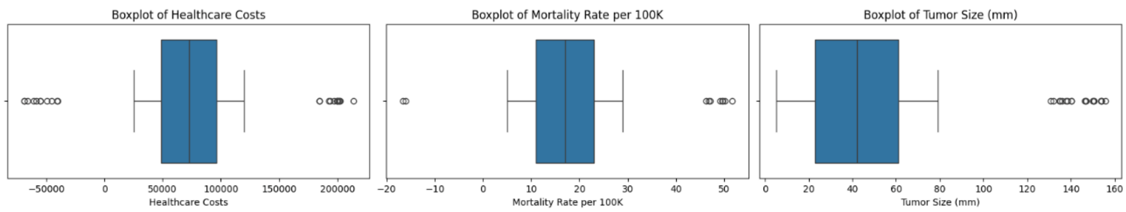

Figure 4 – Bloxplot analysis of ‘Healthcare costs’, ‘Mortality Rate per 100K’ and ‘Tumor Size (mm)’

Regarding numerical variables, features such as `Healthcare Costs`, `Mortality Rate per 100K` and `Tumor Size (mm)` exhibited substantial variability and included several implausible outlier values, including negative figures in cost and mortality rate columns. These were deemed biologically and logically invalid and required correction during preprocessing. Below are the boxplots of the respective variables.

No strong correlations were found among the variables, as all pairwise coefficients were close to zero. In addition, none of the numerical variables showed strong discriminative power with respect to the target variable; their distributions and median values were similar across the groups defined by ‘Survival Prediction’.

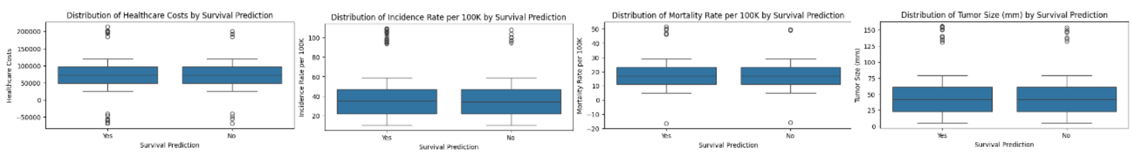

Figure 5 – Boxplot analysis by target

Data Preprocessing and Feature Engineering

During preprocessing, a structured pipeline was implemented to clean, impute, and transform the data. First, unnecessary or invalid columns such as ‘Marital Status’, ‘Diabetes History’ (entirely missing) and 'Transfusion History' (a single constant value) were removed, along with 'Smoking History' due to its redundancy with 'Non Smoker'. Categorical columns with minor missing values, such as 'Healthcare Access' and 'Gender' were imputed using the mode of the valid entries after filtering out noise values like *'?'* for the former or *'P'* for the latter. Rows with *nulls* and duplicate records were removed to maintain data integrity and ensure a clean base for modeling.

Some new features were engineered to enhance the dataset's predictive capacity. The patient's age was derived from the *'Date of Birth'*, *'Advanced_age'* which was created to identify patients aged 65 or older, who may have an increased cancer risk. Additionally, a *'Cardiometabolic Risk'* feature was built by combining the presence of diabetes and heart disease into a single variable reflecting cumulative cardiovascular risk. Finally, the *'Cancer_Severity'* feature integrates Cancer Stage and Tumor Size into a composite index, capturing two key prognostic factors that may improve outcome prediction.

Outliers in numeric features (e.g. *'Healthcare Costs'* and *'Tumor Size'*) were addressed using the interquartile range (IQR) method with Winsorization, where values beyond 1.5 IQRs from the first and third quartiles were "clipped" to the respective bounds. Negative values were imputed with the median of the corresponding variable. All numerical variables were also scaled using Robust Scaler, chosen for its robustness to outliers and skewness, even though the skewness in the data was not substantial. Min-Max scaling was also tested but resulted in less favorable model performance.

Ordinal variables like *'Cancer Stage'*, *'Diet Risk'* and *'Obesity BMI'* were encoded according to their natural order. Nominal categorical variables like *'Country'*, *'Insurance Costs' and 'Treatment Type'* were one-hot encoded. Binary categorical columns, such as 'Yes/No' indicators, were converted to boolean values to ensure model compatibility.

Following these steps, the final dataset was consistently formatted and free of missing or wrong values, with all categorical and continuous data appropriately transformed. This not only ensured data integrity but also enhanced the overall structure and usability of the dataset for subsequent modeling, enabling accurate, interpretable, and generalizable predictions.

Binary Classification

Objective and Strategy

The core task of this project is to develop and evaluate binary classification models capable of predicting the survival of colorectal cancer (CRC) patients. Given the importance of such predictions and the need for generalizability, a rigorous modeling strategy was adopted. It includes:

- Careful feature preprocessing and engineering (as previously described)
- Feature selection techniques for reducing noise and complexity.
- Balanced training/validation splits with stratification.
- Systematic model benchmarking using multiple algorithms

- Use of F1-Macro Score as the main evaluation metric, because it gives equal importance to both precision and recall, in both classes.
- Optimization and hyperparameter tuning to improve model performance

Dataset Preparation and Preprocessing Execution

The modeling pipeline began with the loading and inspection of the dataset, followed by the execution of previously defined preprocessing routines. The dataset, which contained detailed patient-level data, was imported directly from a public repository.

Using the `split_sets()` function, the dataset was divided into **training (80%) and validation (20%)** subsets, ensuring stratification to preserve class distribution. Preprocessing was then applied separately:

- The training set underwent extensive cleaning, encoding, and feature engineering as explained in the Data Preprocessing and Feature Engineering section through the user-defined function `preprocess_train_df()`.
- The validation set was processed using the user-defined function `preprocess_test_df()`, which ensured transformations aligned with the training set (e.g., encoding mappings, medians for imputation).

This resulted in two fully preprocessed datasets (*X_train_processed*, *X_val_processed*) and their corresponding labels, ready for model training and evaluation.

Feature Selection

Given the high-dimensional nature of the dataset after encoding, feature selection was performed to reduce noise, minimize overfitting, and improve model interpretability. Two distinct approaches were used: **Sequential Feature Selection (SFS)** and **Recursive Feature Elimination with Cross-Validation (RFECV)**.

- ***Sequential Feature Selection (SFS)***

Sequential Feature Selection (SFS) is a greedy search algorithm that adds or removes features one at a time based on model performance. Two variants were tested using a logistic regression estimator:

- Forward Selection (SFS-Forward): Begins with no features and adds one at a time.
- Backward Selection (SFS-Backward): Begins with all features and removes the least useful ones iteratively.

Both strategies aimed to retain the top 15 features based on f1_macro score:

a. Selected Features by SFS-Forward:

```
['Alcohol Consumption', 'Cancer Stage', 'Diabetes', 'Diet Risk',
'Early Detection', 'Family History', 'Gender', 'Genetic Mutation',
'Healthcare Access', 'Healthcare Costs', 'Heart Disease History',
'Incidence Rate per 100K', 'Inflammatory Bowel Disease',
```

'Mortality Rate per 100K', 'Obesity BMI']

b. Selected Features by SFS-Backward:

```
['Country_Nigeria', 'Country_Pakistan', 'Country_South Africa', 'Country_South Korea', 'Country_UK', 'Country_USA', 'Insurance Costs_Extended', 'Insurance Costs_No insurance', 'Treatment Type_Combination', 'Treatment Type_Radiotherapy', 'Treatment Type_Surgery', 'Hypertension_Yes', 'Insurance Status_Uninsured', 'Non Smoker_Yes', 'Urban or Rural_Urban']
```

Backward selection mainly retained socioeconomic, health-related, and geographic variables, with country-related features likely serving as proxies that implicitly capture information about the healthcare systems and structural differences across countries. This may reflect disparities in access to care or healthcare quality associated with specific regions or insurance types.

- ***Recursive Feature Elimination with Cross-Validation (RFECV)***

To complement SFS, RFECV was applied using 5-fold cross-validation and a logistic regression classifier as the estimator. This method recursively removes features and evaluates model performance at each step using the f1 metric. The number of features to retain was determined automatically.

a. Final Selected Features by RFECV:

```
['Diabetes', 'Cardiometabolic_Risk', 'Country_Brazil', 'Country_Canada', 'Country_China', 'Country_France', 'Country_Germany', 'Country_India', 'Country_Italy', 'Country_Japan', 'Country_Nigeria', 'Country_Pakistan', 'Country_South Korea', 'Country_USA', 'Treatment Type_Radiotherapy']
```

This selection captures both critical metabolic comorbidities ('*Diabetes*', '*Cardiometabolic_Risk*') and a broad spectrum of geographic variables (represented by 11 distinct countries), reinforcing the notion that country-specific healthcare environments are crucial predictors of survival outcomes. The inclusion of '*Treatment Type_Radiotherapy*' as the sole therapeutic modality highlights the singular importance of this treatment approach in determining patient prognosis.

As we had different alternatives, all with high potential in terms of interpretability, we performed an assessment using a Cross-Validation method for different models to select the set of features that yielded better results and stability. This approach will be explained in more detail in the following subsection.

Model Benchmarking and Comparison

We decided to follow a unified approach for selecting both the best features combination from the applied methods above, and the potential models for obtaining the best possible predictions. For that purpose, we created the `evaluate_models_with_cv()` function that performs a 10-fold Cross-Validation for three models: Logistic Regression, Random Forest, and a Decision

Tree. This way we guarantee we will use the best possible set of features with stable models and obtain proper predictions.

After comparing results with the different alternatives, including an option for keeping all the original features, we decided to continue with the features presented by the SBS backward method:

Model	***Mean F1 Score***	***Score Standard Dev.***	***Overfit Gap***
Random Forest	0.50	0.005	0.019
Logistic Regression	0.50	0.004	0.006
Decision Tree	0.49	0.006	0.009

Table 1 – Models' performance on a 10-fold Cross-Validation

We selected **SBS Backward feature selection** because it demonstrated superior model stability with notably low standard deviations (0.004-0.007 in validation) across all models while maintaining competitive F1 scores (RandomForest reaching 0.502). This method also showed minimal overfit gaps, particularly for LogisticRegression (0.006), indicating strong generalization capabilities crucial for medical applications where prediction consistency is as important as performance.

Hyperparameter Tuning and Final Model Selection

After an initial evaluation, we selected **Logistic Regression** and **Random Forest** as our primary candidates as both demonstrated good performance and stability across cross-validation folds. For Decision Trees, despite their interpretability and decent results, theory suggests they often show a clear tendency to overfit, so we decided to rule this model out and try with ensemble methods such as AdaBoost, Voting Classifier, and Stacking Classifier. These new approaches combine the strengths of multiple base learners to improve predictive performance and robustness.

Excepting Voting and Stacking Classifiers, with the chosen potential models, we performed a Grid Search with a 5-fold cross-validation to find the best hyperparameter combination for each candidate. Given preliminary attempts that resulted in models only predicting the positive class, we opted to balance class weights for all candidates (when possible), aiming to address the slight class imbalance (approximately 60% positive vs 40% negative) and to minimize false positives, which we consider more costly in this oncology prediction context since incorrectly predicting survival could lead to insufficient treatment intensity or premature termination of necessary care.

For our optimization metric, we chose **F1-macro instead of regular F1**. Even though F1-macro is usually used for problems with multiple classes, we found it worked well for our binary classification project. It gives the same importance to both classes, which helped our models predict better the negative class, instead of just focusing on positive predictions. This was especially useful when our models initially tended to only predict the positive class. Beyond F1-

macro, we also evaluated the model's using precision, recall, ROC-AUC curves, and examined their overfit gap (difference between training and validation performance).

Note that while we used cross-validation for our initial model selection process, we ultimately used a holdout validation set (20% of our original data) to make our final model assessment, providing a more realistic estimate of how our models would perform on completely unseen data.

Model	***F1 Macro***	***F1 Standard***	***Precision***	***Recall***	***Specificity***	***AUC***	***Overfit Gap***
Logistic Regression	0.50	0.58	0.60	0.56	0.43	0.501	0.006
Random Forest	0.50	0.56	0.61	0.52	0.48	0.504	0.04
AdaBoost	0.50	0.60	0.60	0.61	0.39	0.498	0.01
Stacking Classifier	0.50	0.56	0.60	0.52	0.49	0.504	0.04
Voting Classifier	0.50	0.59	0.60	0.57	0.43	0.501	0.01

Table 2 – Models' performance on a validation set (hold out)

Ultimately, we ended up selecting the **Voting Classifier** as our final model because it gave the best balance between reducing false positives and maintaining stable predictions. With a precision of 0.60 and minimal overfitting (gap of 0.01), it works consistently on both training and test data. While Random Forest has slightly better precision (0.61) and Stacking has better specificity (0.49), both tend to overfit more with gaps of 0.04, making them less reliable for new patients.

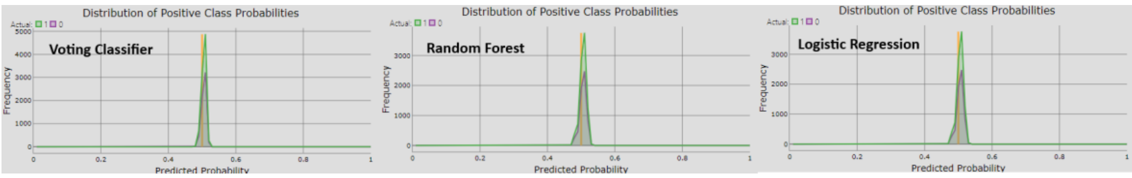

Figure 6 – Distribution of class probabilities for different models

The Voting Classifier combines the strengths of Logistic Regression and AdaBoost using weights of 3 for the former, and 2 for the latter, giving us better recall (0.57) without losing precision. All our models have AUC values around 0.50, **showing the difficulty of separating the survival classes** with our current features *(see Figure 6)*, but the Voting Classifier still maintains good specificity (0.43) and produces fewer false positives than AdaBoost. This makes it our best option when we need reliable predictions and want to minimize false positives, even with the existent limitations.

Final Test Set Prediction and Kaggle Submission

After final model selection, predictions were generated for the test dataset provided by the competition organizers. The pipeline used for preprocessing the test set was identical to that used for the training data, ensuring full consistency.

The model was trained using the Voting Classifier model and used to predict survival on the test set. The predictions were formatted as required for Kaggle submission. The selected submission for the final assessment resulted in a F1-Weighted public score of 0.51973. Although this metric is not explicitly mentioned in this report, the obtained public score is similar to the F1-Weighted score obtained for the validation set using the Voting Classifier model, approximately 0.52.

Open-Ended Section

Feature selection highlighted several impactful features, especially country-specific ones. This prompted additional investigation of how differences in healthcare systems, treatment access and other contextual factors across countries might influence survival, particularly in the context of CRC. Let's explore each country.

Nigeria's healthcare system faces many limitations, particularly in oncology infrastructure and diagnostic capacity. A larger number of CRC cases are diagnosed at an advanced stage, often needing emergency care. Studies have shown that emergency cases are associated with significantly reduced survival, with a median overall survival of around 6.4 months compared to 17.4 months in elective cases (i.e. patients who are diagnosed with and treated in a planned, non-emergency setting). The high preoperative mortality (i.e. the rate of deaths occurring during the period encompassing the preoperative period, the intra-operative period (during surgery) and the postoperative period) underlines the critical need for better early detection and coordinated treatment strategies.

In **Pakistan**, CRC frequently affects individuals below the age of 50 and patient outcomes are highly influenced by biomarkers, such as high levels of CEA (carcinoembryonic antigen >5 ng/ml)[2], and the availability of surgical options. Those who do not receive surgery face much higher mortality risks. The absence of national screening initiatives and limited access to advanced therapies contribute to lower survival rates.

The outcomes for CRC in **South Africa** are heavily dependent on the progression of the disease. Although localized cases have a relatively strong 5-year survival rate of ~72%, the survival rate drops significantly when metastases are present - ~57% with liver metastases and only ~26% when the liver and lung are involved. While there is no direct link between CRC and hypertension in South Africa, studies show that hypertension is a known contributor to postoperative complications, potentially worsening outcomes. On the other hand, surgical access and the use of chemotherapy have begun to close some of the gaps regarding the determinants of survival.

In contrast to the countries mentioned above, **South Korea** represents a model of effective CRC management, largely due to a robust national screening program and universal healthcare coverage. Between 1996 and 2015, 5-year survival increased from ~59% to 75%. The results are particularly favorable for cancers of the left colon, due to early diagnosis and timely intervention. Although hypertension increases CRC by ~17%, South Korea's disease management infrastructure mitigates a lot of its adverse impact on outcomes. Treatments like chemotherapy and surgery are the key drivers of the country's survival statistics.

[2] Carcinoembryonic antigen (CEA) is a protein involved in cell adhesion that is normally present at low levels but can rise significantly in certain cancers, making high levels a potential sign of tumor growth or recurrence.

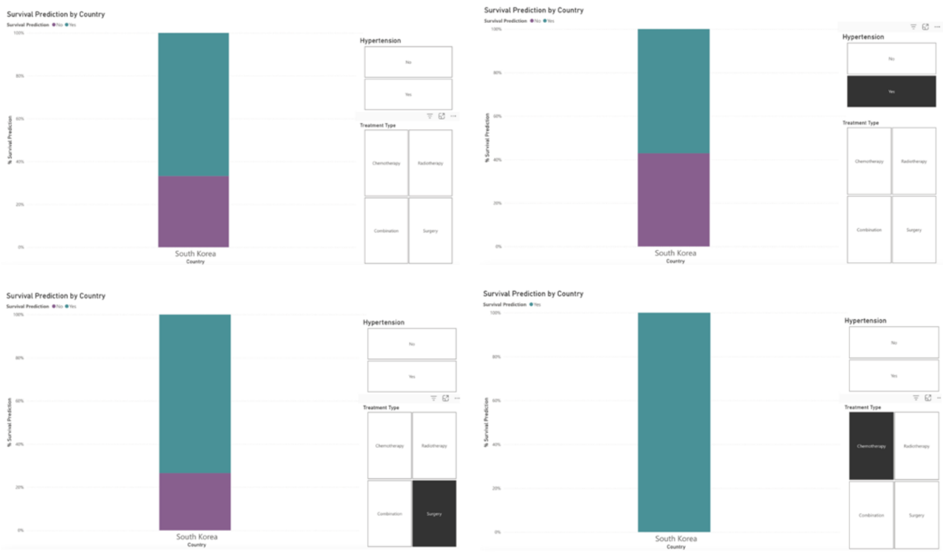

Figure 7 – Survival Prediction in South Korea, filtered by Hypertension and Treatment Type

England's National Health Service (NHS) provides uniform screening and cancer care. Prognosis varies by stage at diagnosis, with stage I CRC showing about 90% 5-year survival, stage II 85%, stage II 65%, and stage IV falling to 10%. Universal access to medical services has substantially contributed to improved long-term outcomes. The presence of untreated hypertension has been linked to a modest increase in CRC risk effect on mortality (~12%), but effective public health management through the NHS largely neutralizes its effect on mortality.

Despite advanced medical technology and innovative treatments, CRC outcomes in the US are influenced by disparities in insurance coverage. Individuals without comprehensive insurance often experience late screening and inadequate treatment, which affects their survival. While 5-year survival for localized CRC is 91%, this rate drops to 13% in cases diagnosed at a more advanced stage. Interestingly, the use of anti-hypertensive medication has shown a decrease in CRC mortality, though this benefit is mainly observed in insured populations with better access to both hypertension management and cancer. Chemotherapy and surgery offer the highest survival benefits, but access remains a critical determinant.

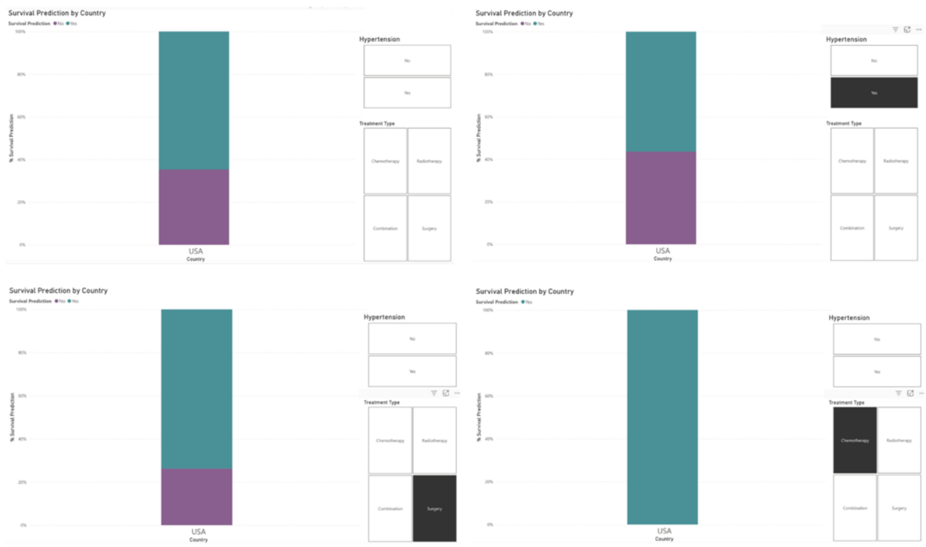

Figure 8 – Survival Prediction in USA, filtered by Hypertension and Treatment Type

As can be observed in the image below, the visualized survival predictions are consistent with the findings presented in the preceding research. This alignment suggests that the countries highlighted in the analysis serve indeed as reliable indicators for predicting CRC patient survival.

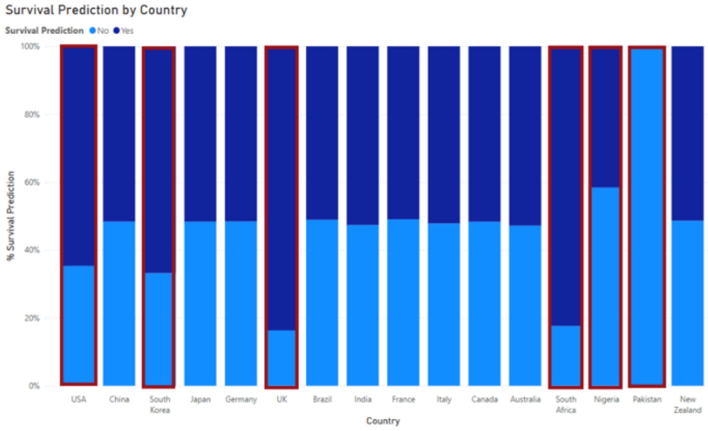

Figure 9 – Survival Prediction by Countries

On the other hand, another variable highlighted by feature selection was `Urban or Rural_Urban`. CRC mortality is consistently higher in rural areas than in urban areas. Rural regions show a 3.5 percentage point increase in overall mortality and a 1.9 percentage point rise in CRC-specific mortality. This disparity is the result of multiple factors: lower screening rates, delayed diagnosis, and treatment due to fewer healthcare resources, socio-economic challenges, and transportation barriers.

Country-specific data highlights the extent of this divide. In Nigeria, rural populations face higher mortality due to poor access to screening and treatment, with urban centers only relatively better equipped. In Pakistan, cities like Karachi and Laore offer oncology services, while rural areas lack basic infrastructure, leading to higher rural mortality. South Africa sees similar trends, with urban residents accessing private care and rural populations relying on under-resourced public services.

Countries with universal healthcare show narrow gaps. South Korea's national screening reduces disparities, although rural areas still fall slightly behind. In the UK, the NHS ensures access, but rural patients face longer travel times and delayed diagnoses, resulting in higher mortality.

In the US, disparities are accentuated and increasing. Rural Americans experience higher mortality and slower declines in death rates, with rural counties reporting 180 deaths per 100000 inhabitants compared to 158 in urban counties. Lower adherence to screening and limited access to healthcare are the main causes of this growing gap.

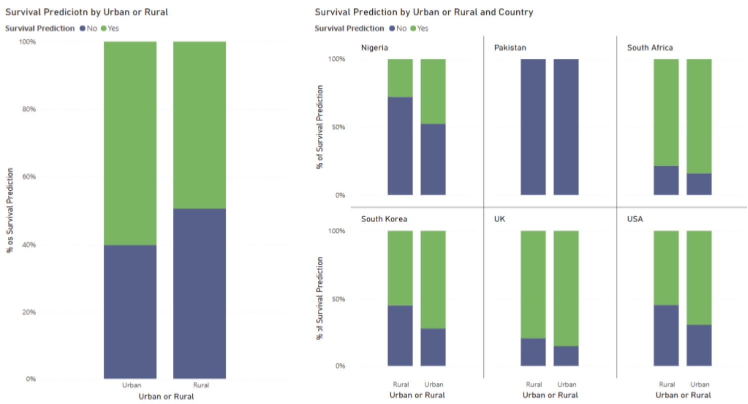

Figure 10 – Survival Prediction by Urban or Rural

As can be seen in the graphics above, the data broadly supports the findings mentioned, with urban areas generally showing higher survival rates. While not perfectly aligned – particularly in Pakistan, where outcomes are uniformly poor - the overall trend remains consistent with the research.

Conclusion

The primary objective of this project was to develop a predictive model for colorectal cancer (CRC) survival by integrating clinical, demographic, and socioeconomic variables. Through comprehensive data preprocessing, targeted feature engineering, and systematic model benchmarking, multiple machine learning algorithms were evaluated and compared.

Following feature selection, several models—including Logistic Regression, Random Forest, AdaBoost, and Stacking—were trained, optimized, and validated using stratified cross-validation. The Voting Classifier, combining Logistic Regression and AdaBoost with weights of 3 and 2, respectively, was ultimately selected as the final model. This ensemble achieved a precision of 0.60, recall of 0.57, and demonstrated minimal overfitting with a validation gap of only 0.01, ensuring stable performance across both training and validation sets. While Random Forest and Stacking exhibited slightly higher precision or specificity, they also showed greater overfitting, reducing their reliability on unseen data. Although class separability remained limited across all models (AUC values near 0.50), the Voting Classifier provided the most balanced and reliable predictions, while also minimizing false positives, which is critical for clinical applicability.

The feature selection process also provided valuable insights into factors influencing patient survival. In addition to core clinical predictors such as cancer stage, age, treatment type, early detection, and healthcare access, several systemic factors—such as country of origin, urban versus rural residence, and health insurance coverage—emerged as strong predictors. These findings emphasize that CRC outcomes are shaped not only by biological characteristics but also by broader social determinants and healthcare infrastructure. Countries with well-established healthcare systems, such as South Korea and the United Kingdom, were consistently associated with higher survival rates, while countries with more limited resources, such as Nigeria and Pakistan, were associated with poorer outcomes. Urban residency similarly correlated with improved survival, further reinforcing the model's ability to capture real-world disparities in healthcare delivery.

Despite these promising results, some limitations must be acknowledged. The dataset presented inherent imbalances and possible inconsistencies. Additionally, some variables, particularly socioeconomic indicators, may not have been captured with the same level of detail across all countries or regions, which could introduce bias. The generalizability of the model to external populations remains uncertain.

Future work may focus on incorporating more granular data sources, such as genetic profiles, treatment adherence, or comorbidity indices, to improve predictive power. Alternative ensemble approaches that balance performance with interpretability could also be explored. Furthermore, validating the model on external cohorts or in clinical practice would provide a more robust assessment of its real-world applicability. From a public health perspective, the model could serve as a valuable tool for simulating the impact of interventions aimed at improving healthcare access and screening programs, particularly in resource-limited settings.

In summary, this project developed a functional survival prediction model for colorectal cancer patients, while also uncovering meaningful patterns with clinical and public health implications. By combining clinical, demographic, and socioeconomic dimensions, the model offers a comprehensive view of CRC survival, highlighting both medical and structural factors that can be addressed to improve patient outcomes worldwide.

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